

JavaScript Loops – Premium Notes

Complete notes for For Loop, While Loop, and Do While Loop with syntax, examples, patterns, and interview tips.

1. Quick Comparison

Loop	Best Use	Checks Condition	Runs Minimum
for	Known repetitions	Before execution	0 times
while	Unknown repetitions	Before execution	0 times
do while	Run once then repeat	After execution	1 time

For Loop

Syntax

```
for(let i=1; i<=5; i++){  
  console.log(i);  
}
```

Use Cases

Counting, arrays, tables, patterns.

Examples

Print 1 to 10
Even numbers
Factorial
Nested star patterns

While Loop

Syntax

```
let i=1;  
while(i<=5){  
  console.log(i);  
  i++;  
}
```

Use Cases

When number of repetitions is unknown.

Examples

Menu systems, input validation, countdown.

Do While Loop

Syntax

```
let i=1;  
do{  
  console.log(i);  
  i++;  
}while(i<=5);
```

Use Cases

Run code at least once.

Examples

Login attempt, ask-again menus.

2. Important Programs

1) Sum of 1 to 10 using for:
`let sum=0;
for(let i=1;i<=10;i++){ sum+=i; }`

2) Factorial using while:
`let i=1, fact=1;
while(i<=5){ fact*=i; i++; }`

3) Table using do while:
`let i=1;
do{
 console.log(5*i);
 i++;
}while(i<=10);`

3. Break and Continue

break → stops loop immediately.

continue → skips current iteration and moves next.

4. Nested Loops Pattern

```
for(let i=1;i<=5;i++){  
  let row='';  
  for(let j=1;j<=i;j++){ row+='* '; }  
  console.log(row);  
}
```

5. Interview Questions

- Difference between while and do while?
- Which loop is best for array traversal?
- What is infinite loop?
- What do break and continue do?

6. Final Tips

Use for loop when count is known, while when condition-based repetition is needed, and do while when code must execute once.